

Testing and Evaluation

The AES Vault system was evaluated to determine whether the encryption engine performed correctly under normal operating conditions as well as under failure scenarios. The purpose of the testing phase was to validate that the system could encrypt files securely, restore the original content with the correct password, reject invalid inputs, detect tampered ciphertext, and prevent accidental overwriting of existing destination files.

The project was assessed using automated test cases designed specifically for the cryptographic functions. These tests were executed against the encryption and decryption logic to verify the accuracy, reliability, and security of the implementation. The evaluation focused on the functional correctness of the AES-256-GCM encryption mechanism and on the protection of the system against invalid user input and file tampering.

The test suite was executed using the following command:

```
cd "c:\Users\maral\Downloads\SEM 3\Cryptography\Project\Code"; &  
"C:/Users/maral/AppData/Local/Programs/Python/Python313/python.exe" -m pytest -q
```

The output of the test execution was:

5 passed in 0.10s

This result indicates that all five evaluation tests passed successfully. The tests covered the following areas:

- Round-trip encryption and decryption: the system successfully encrypted a file and restored the original data without loss.
- Wrong password rejection: the system correctly rejected incorrect passwords and prevented unauthorized decryption.
- Tampered ciphertext detection: modified encrypted files were identified and rejected.
- Malformed file validation: invalid encrypted input was rejected without causing system failure.
- Destination overwrite protection: the system prevented accidental overwriting of existing files.

Based on the test results, it can be concluded that the AES Vault encryption system functions correctly and meets the expected functional and security requirements. The implementation successfully handled valid encryption, secure decryption, password validation, tamper detection, and invalid input handling, confirming that the testing and evaluation phase was successful.